Laxmi Charitable Trust’s

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Full Stack development MERN Practical

Practical – 2

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Class: TYIT	Batch: 01
Date of Assignment: 30/06/2026	Date/Time of Submission: 30/06/2026

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code (Calculate Salary with Bonus) :

```
function calculateSalary(name, basicSalary)
{
let bonus = 5000;
let totalSalary = basicSalary + bonus;

console.log("Employee Name = ", name);
console.log("Bsaic Salary = ", basicSalary);
console.log("Bonus = ", bonus);
console.log("Total salary = ", totalSalary);
}
calculateSalary("vinayak padhy", 50000);
```

Output :

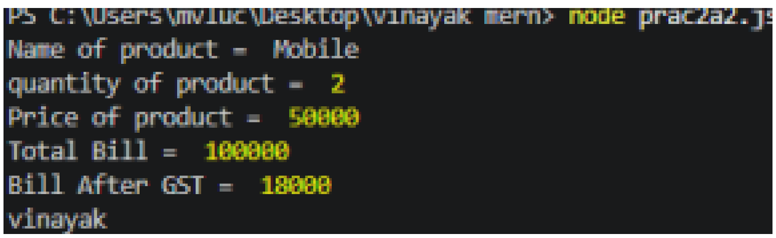
```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2a.js
Employee Name =  vinayak padhy
Bsaic Salary =  50000
Bonus =  5000
Total salary =  55000
PS C:\Users\mvluc\Desktop\vinayak mern>
```

Code (Calculate Product Price with 18% GST) :

```
function genBill(product, quantity, price){
```

```
 let total = quantity*price;
 let Gst = total*0.18;
 console.log("Name of product = ", product);
console.log("quantity of product = ",quantity);
console.log("Price of product = ", price);
console.log("Total Bill = ", total);
 console.log("Bill After GST = ", Gst);
}
genBill("Mobile",2,50000);
console.log("vinayak");
```

Output:-



```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2a2.js
Name of product =  Mobile
quantity of product =  2
Price of product =  50000
Total Bill =  100000
Bill After GST =  18000
vinayak
```

Output :

b. Work with arrays and objects in JavaScript.

Code :

```
let fruits = ["Mango","Banana","Apple"];

let student = {
 name: "vinayak",
 age: 19,
 city: "Mumbai"
};

console.log("Fruits :",fruits);
console.log("Second Fruits :",fruits[1]);

console.log("Student Name:", student.name);
console.log("Student Age: ",student.age);
console.log("Student City:", student.city);
```

Output :

```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2b.js
Fruits : [ 'Mango', 'Banana', 'Apple' ]
Second Fruits : Banana
Student Name: vinayak
Student Age:  19
Student City: Mumbai
```

c. Demonstrate use of arrow functions.

Code :

```
let add = (a,b) => {
 return a+b;
};
let sub = (a,b) => {
 return a-b;
};
let mul = (a,b) => {
 return a*b;
};
console.log("Sum = ",add(10,20));
console.log("Sub = ",add(10,20));
console.log("Mul = ",mul(5,9));
console.log("vinayak");
```

Output :

```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2c.js
Sum =  30
Sub =  30
Mul =  45
vinayak
```

d. Write a program using array methods (map, filter, reduce).

Code :

```
let numbers = [10, 20, 30, 40, 50];

// Map: double each number
let doubled = numbers.map(num => num * 2);

// Filter: numbers greater than 25
let greater25 = numbers.filter(num => num > 25);
```

```
// Reduce: sum of all numbers
let total = numbers.reduce((sum, num) => sum + num, 0);
console.log("Original Array: ", numbers);
console.log("Map: ", doubled);
console.log("Filter: ", greater25);
console.log("Reduce: ", total);
console.log("vinayak");
```

Output :

```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2d.js
Original Array:  [ 10, 20, 30, 40, 50 ]
Map:  [ 20, 40, 60, 80, 100 ]
Filter:  [ 30, 40, 50 ]
Reduce:  150
vinayak
```

e. Create a program that manipulates objects and displays output. **Code :**

```
//Creating an Object
let student ={
 rollNo:112,
 name:"Suyash KArnale",
 marks: 90,
};
console.log("Original Object:");
console.log(student);

// 2. Accessing Properties
console.log("\nAccessing Properties: ");
console.log("Name:", student.name);
console.log("Marks:",student.marks);

//Updating Properties
student.marks = 95;
console.log("\nAfter Updating Marks:");
console.log(student);

//4.Adding New Properties
student.city = "Mumbai";
console.log("\nAfter Adding City:");
```

```
console.log(student);

//5. Deleting a Property
delete student.rollNo;
console.log("\nAfter Deleting Roll No:");
console.log(student);

console.log("vinayak");
```

Output :

```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2e.js
Marks: 90

After Updating Marks:
{ rollNo: 112, name: 'Suyash KArnale', marks: 95 }

After Adding City:
{ rollNo: 112, name: 'Suyash KArnale', marks: 95, city: 'Mumbai' }

After Deleting Roll No:
{ name: 'Suyash KArnale', marks: 95, city: 'Mumbai' }
vinayak
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code :

```
//Array of objects
let students = [
 {
 rollNo: 103,
 name: "Suyash",
 marks: 90,
 },
 {
 rollNo: 104,
 name: "Rahul",
 marks: 85,
 },
 {
 rollNo: 105,
```

```
 name: "Priya",
 marks: 95,
 }
];
//Finction to display student details
function display(s)
{
 console.log("student details");
 console.log(`--------------------`);
for(let stud of s)
 {
 console.log(`Roll No: , ${stud.rollNo}`);
console.log(`Name: , ${stud.name}`);
console.log(`Marks: , ${stud.marks}`);
console.log(`--------------------`);  }
}
//Function Call
display(students);
console.log("vinayak");
```

Output :

```
PS C:\Users\mvluc\Desktop\vinayak mern> node prac2f.js
Marks: , 90
--------------------
Roll No: , 104
Name: , Rahul
Marks: , 85
--------------------
Roll No: , 105
Name: , Priya
Marks: , 95
--------------------
vinayak
```